

Project Context:

- Climate change is worsening due to fossil fuel dependence and human activities (deforestation, waste management, transportation) leading to more extreme weather conditions (extreme heat, typhoons).
- Actions made by the government are not enough, such as websites that contain information about climate change and its impacts.
- There is a lot of available energy and climate data available, but mostly are difficult for the public to understand.

Hypothesis:

- **Null Hypothesis:** Lumi does not significantly improve the ability of the users to understand and evaluate their renewable energy options
- **Alternative Hypothesis:** Lumi significantly improve the ability of the users to understand and evaluate their renewable energy options

Purpose:

- Help users to **understand** and **evaluate** their options in renewable energy depending on their location while also showing them the costing (installation, materials needed) through valid supporting data.
- Enhance their **awareness** of climate change and its impacts.

Description (System Features):

- **Landing page: Home**
 - Short description of the system
 - Button for “learn more” proceeds to About section
- **Information about the system: About**
 - Current problem supported by data (source:)
 - Climate change impacts
 - Limited awareness of people about renewable energy
 - Purpose of Lumi
 - 1
 - 2
 - 3
 - 4
 - Key features (includes uses) and beneficiaries
 - Data sources
- **Climate and Energy and Regional Insights dashboard: EnergyHub**
 - Interactive dashboard for climate and energy analysis
 - Displays temperature trends, rainfall patterns, energy consumption, and electricity demand

- Integrates data from NGCP and OpenWeather API
- Allow users to filter and navigate data by:
 - Luzon
 - Visayas
 - Mindanao
 - Specific region or cities (depending on data availability)
- Provides AI-generated interpretations for every chart and visualization
- Includes climate change awareness insights and regional observations
- Shows renewable energy suitability summaries per region
- Uses visual analytics such as:
 - KPIs
 - Trend charts
 - Comparative graphs
 - Regional overview panels

- **Predictive Forecasting Module: Forecasting**

- Uses historical climate and energy data for forecasting
- Predicts future energy demand and consumption trends
- Applies trend analysis and forecasting techniques
- Provides AI-generated explanations and interpretations for forecasts
- Helps users identify:
 - Future energy requirements
 - Possible energy shortages
 - Climate-related energy impacts
- Supports proactive planning and decision-making
- Includes forecasting visualizations such as:
 - Line graphs
 - Trend projections
 - Demand forecasting charts

- **Renewable Energy Recommendation and Simulation Engine: Ecosim**

- Recommends the most suitable renewable energy source per region
- Uses a rule-based recommendation system combined with AI analysis
- Compares renewable energy sources such as:
 - Solar
 - Hydro
 - Wind
- Compares renewable sources with current traditional energy sources
- Allows users to input monthly electricity consumption or bills
- Simulates:
 - Estimated costs
 - Potential savings
 - Energy efficiency improvements
 - Environmental impacts
- Generates AI-powered comparative summaries and recommendations
- Helps users evaluate renewable energy feasibility and sustainability

- **Contact information: Contact**

- Contact details

- Dedicated email for the system
 - Contact number
 - Location
- **Decision Support Interface:**
 - Responsive and user-friendly interface
 - Compatible with:
 - Smartphones
 - Laptops
 - Tablets
 - Designed for easy navigation and accessibility
 - Supports data-driven decision-making through:
 - Interactive charts
 - AI-generated insights
 - Forecasting tools
 - Renewable energy comparisons
 - Enhances user understanding of climate and energy information
 - Provides a centralized platform for climate monitoring and energy planning

General Objective:

- To develop a web-based environmental intelligence system that provides accessible, data-driven insights (costing included), predictive analytics to help users understand and evaluate their options in renewable energy and to enhance their awareness of climate change and its impacts .

Specific Objective:

- To design and develop a dashboard that visualizes climate patterns and energy trends interpretation;
- To develop a rule-based recommendation engine for identifying suitable renewable energy sources;
- To implement a predictive analytics module for forecasting energy demand trends;
- To create a simple what-if simulation tool for comparing renewable energy scenarios;
- To develop a user-friendly interface that supports data interpretation and decision-making.

Scope:

- Publicly available data - Climate and energy
- Solar, wind, hydro
- Predictive analytics
- Dashboards
- Simulation
- Region selection
- User-friendly outputs

Limitations:

- Publicly available datasets
- Predictive analytics - basic statistical methods

- Rule-based recommendation - should also include the costing
- Solar, wind, hydro

Significance of the Study:

- Students and educators - educational resource
- Households and communities - practical insights
- Government agencies (DOE, DENR, LGUs) - decision support, but will not be used as a professional decision making tool
- Researchers and developers (Computer Science, Environmental Science, Data Analytics) - groundwork for environmental intelligence systems